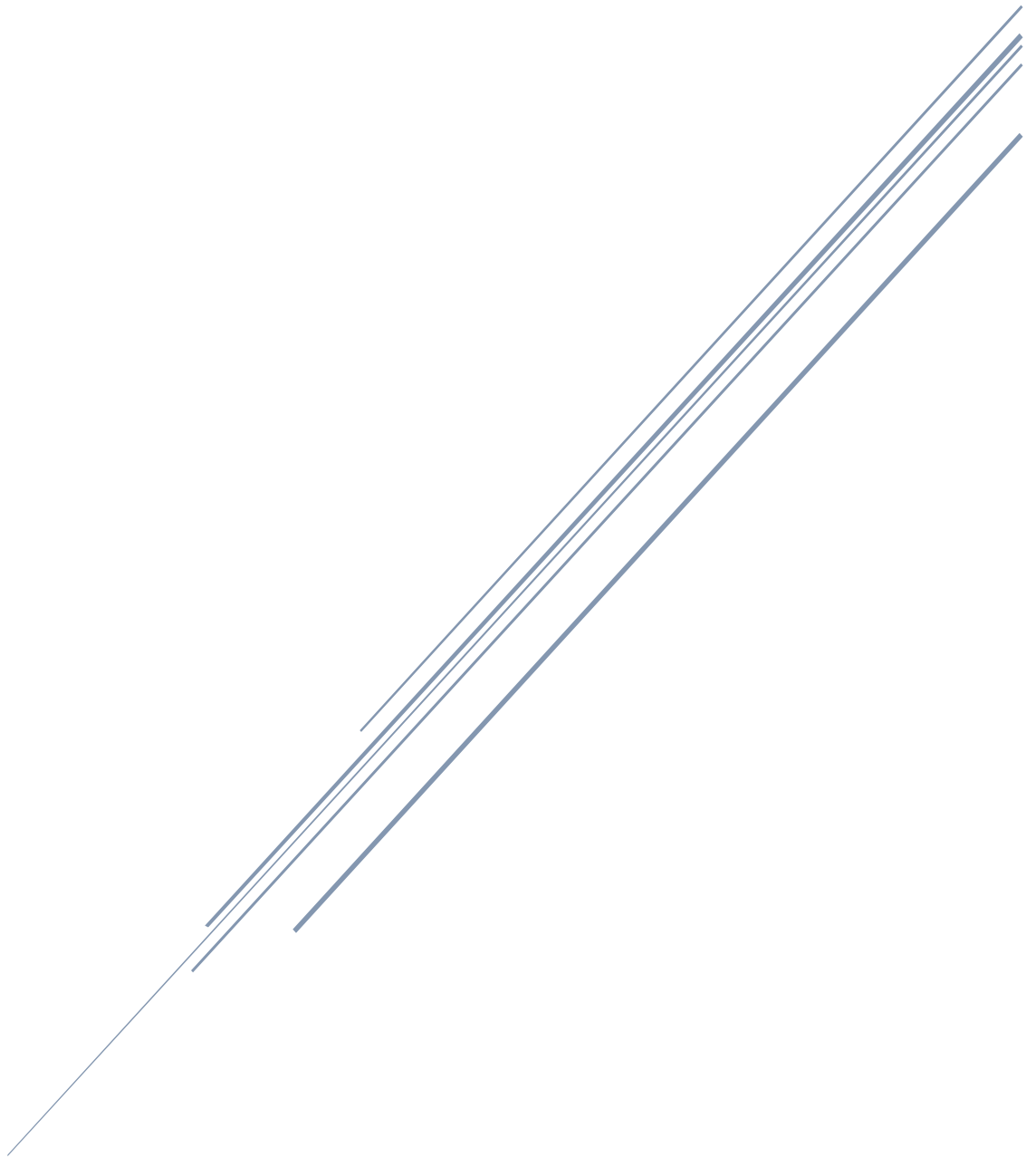


CUSTOMER CHURN ANALYSIS

USING SQL AND POWER BI



SHYAM SUDHEER

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INTRODUCTION

Customer churn refers to the phenomenon where customers stop using a company's product or service over a given period. In industries such as telecommunications, retaining existing customers is significantly more cost-effective than acquiring new ones. Therefore, understanding the reasons behind customer churn is a critical business problem.

This project focuses on analysing customer churn behaviour using a telecom dataset. The objective is to identify patterns, trends, and key factors that influence customer attrition. By leveraging SQL for data analysis and Power BI for visualization, the project aims to provide actionable insights that can help businesses reduce churn and improve customer retention strategies.

The analysis is designed to simulate a real-world business scenario where companies rely on data-driven decision-making to enhance customer experience and maximize profitability.

OBJECTIVES

The main objectives of this project are:

- To calculate the overall customer churn rate
- To identify key factors influencing churn behaviour
- To analyse customer segments based on risk levels
- To examine the combined effect of multiple variables on churn
- To create an interactive dashboard for business insights
- To provide data-driven recommendations for reducing churn

3. DATASET DESCRIPTION

The dataset used in this project is the **Telco Customer Churn dataset**, which contains information about customers and their subscription details.

Key Variables:

- **CustomerID:** Unique identifier for each customer
- **Tenure:** Number of months the customer has stayed with the company
- **Monthly Charges:** Monthly amount charged to the customer
- **Total Charges:** Total amount paid by the customer
- **Contract:** Type of contract (Month-to-month, One year, Two year)

- **Payment Method:** Mode of payment (Electronic check, Credit card, etc.)
- **Internet Service:** Type of internet service
- **Churn:** Indicates whether the customer left (Yes/No)

These variables are used to understand customer behaviour and identify patterns associated with churn.

4. METHODOLOGY

4.1 Data Loading

The dataset was imported into MySQL using the Table Import Wizard. A structured table named churn_data was created to store all customer records.

4.2 Data Cleaning

Data cleaning was performed to ensure accuracy and reliability:

- Missing values in the TotalCharges column were identified and replaced with NULL
- Data type of TotalCharges was converted from text to numeric
- A new column churn_flag was created (1 = churn, 0 = no churn)
- Data consistency checks were performed on categorical variables

This step ensured that the dataset was ready for analysis without errors.

4.3 Feature Engineering

New variables were created to enhance analysis:

- Tenure Group:
 - New (< 12 months)
 - Medium (12–24 months)
 - Long-term (> 24 months)
- Charge Group:
 - Low (< 50)
 - Medium (50–80)
 - High (> 80)

These groupings helped simplify analysis and identify patterns more effectively.

4.4 Data Analysis (SQL)

SQL was used to perform detailed analysis, including:

- Overall churn rate calculation
- Churn rate by contract type
- Churn rate by tenure group
- Churn rate by monthly charges
- Churn rate by payment method
- Combined analysis (contract + charge group)

These queries helped identify key drivers of churn.

4.5 Data Visualization (Power BI)

Power BI was used to create an interactive dashboard:

- KPI cards for total customers, churn count, and churn rate
- Bar charts to compare churn across different categories
- Matrix heatmap to analyze combined factors
- Clean layout and formatting for better readability

The dashboard allows users to easily interpret insights and identify high-risk segments.

5. EXPLORATORY DATA ANALYSIS

5.1 Overall Churn Rate

The overall churn rate was calculated to understand the general level of customer attrition. The dataset shows an approximate churn rate of **27%**, indicating a significant portion of customers leaving the service.

5.2 Churn by Contract Type

Analysis shows:

- Month-to-month contracts have the highest churn rate
- Long-term contracts (one year, two year) have lower churn

Insight:

Customers with short-term commitments are more likely to leave.

5.3 Churn by Tenure

- New customers have the highest churn rate
- Long-term customers show lower churn

Insight:

Customer experience in the early stages plays a critical role in retention.

5.4 Churn by Monthly Charges

- Customers with high monthly charges show higher churn
- Lower charges correspond to lower churn

Insight:

Pricing is a major factor influencing customer decisions.

5.5 Churn by Payment Method

- Customers using electronic check show higher churn
- Customers using automatic payment methods show lower churn

Insight:

Convenience and payment automation reduce churn.

5.6 Combined Analysis (Key Insight)

The most important finding:

- Customers with **month-to-month contracts and high charges** show the highest churn rate

Insight:

Churn is driven by a combination of multiple factors rather than a single variable.

6. DASHBOARD OVERVIEW

Customer Churn Analysis Dashboard

7K

Total Customers

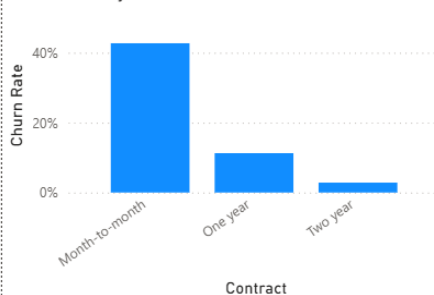
2K

Total Churn

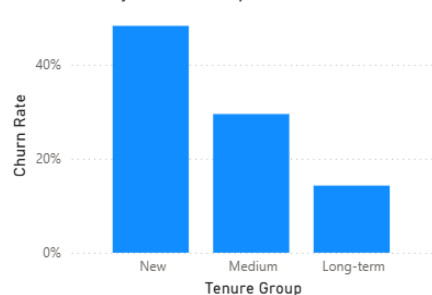
26.54%

Churn Rate

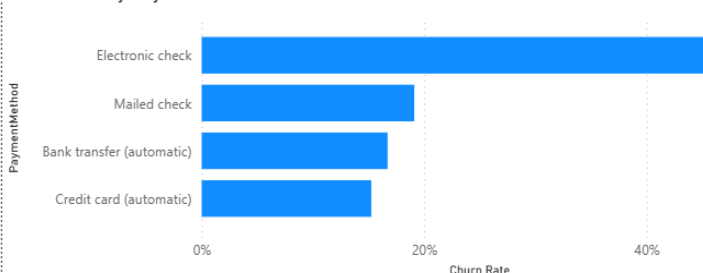
Churn Rate by Contract



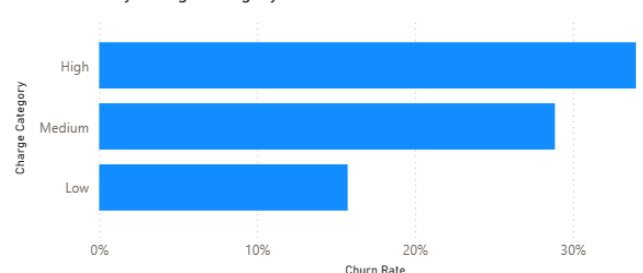
Churn Rate by Tenure Group



Churn Rate by PaymentMethod



Churn Rate by Charge Category



Contract	High	Low	Medium	Total
Two year	5.19%	0.98%	2.31%	2.83%
One year	19.40%	4.03%	8.53%	11.27%
Month-to-month	52.04%	30.18%	42.84%	42.71%
Total	33.99%	15.74%	28.86%	26.54%

Highest Risk Segment: Month-to-month customers with high monthly charges show the highest churn rate (~52%).

The Power BI dashboard includes:

- KPI cards:
 - Total Customers
 - Total Churn
 - Churn Rate
- Visualizations:
 - Churn by Contract Type
 - Churn by Tenure Group
 - Churn by Charges
 - Churn by Payment Method
- Heatmap:
 - Contract vs Charge Group

The dashboard provides a comprehensive view of customer behaviour and risk segmentation.

7. KEY FINDINGS

1. Month-to-month contracts have the highest churn rate
2. New customers are more likely to leave
3. High monthly charges increase churn probability
4. Payment method influences customer retention
5. Combined factors significantly impact churn

8. BUSINESS RECOMMENDATIONS

Based on the analysis:

1. **Improve Customer Retention**
 - Offer incentives for long-term contracts
 - Focus on improving onboarding experience
2. **Pricing Strategy**
 - Review pricing for high-risk customer segments
 - Provide discounts or bundled plans
3. **Payment Optimization**
 - Encourage automatic payment methods
 - Reduce friction in payment processes

4. Targeted Marketing

- Identify high-risk customers early
- Offer personalized retention strategies

9. CONCLUSION

This project demonstrates how data analytics can be used to understand customer churn behaviour and support business decision-making. The analysis highlights that churn is influenced by multiple factors, including contract type, tenure, pricing, and payment methods.

By identifying high-risk segments and applying targeted strategies, companies can reduce churn, improve customer satisfaction, and increase profitability.

The combination of SQL for analysis and Power BI for visualization provides a powerful approach for solving real-world business problems.